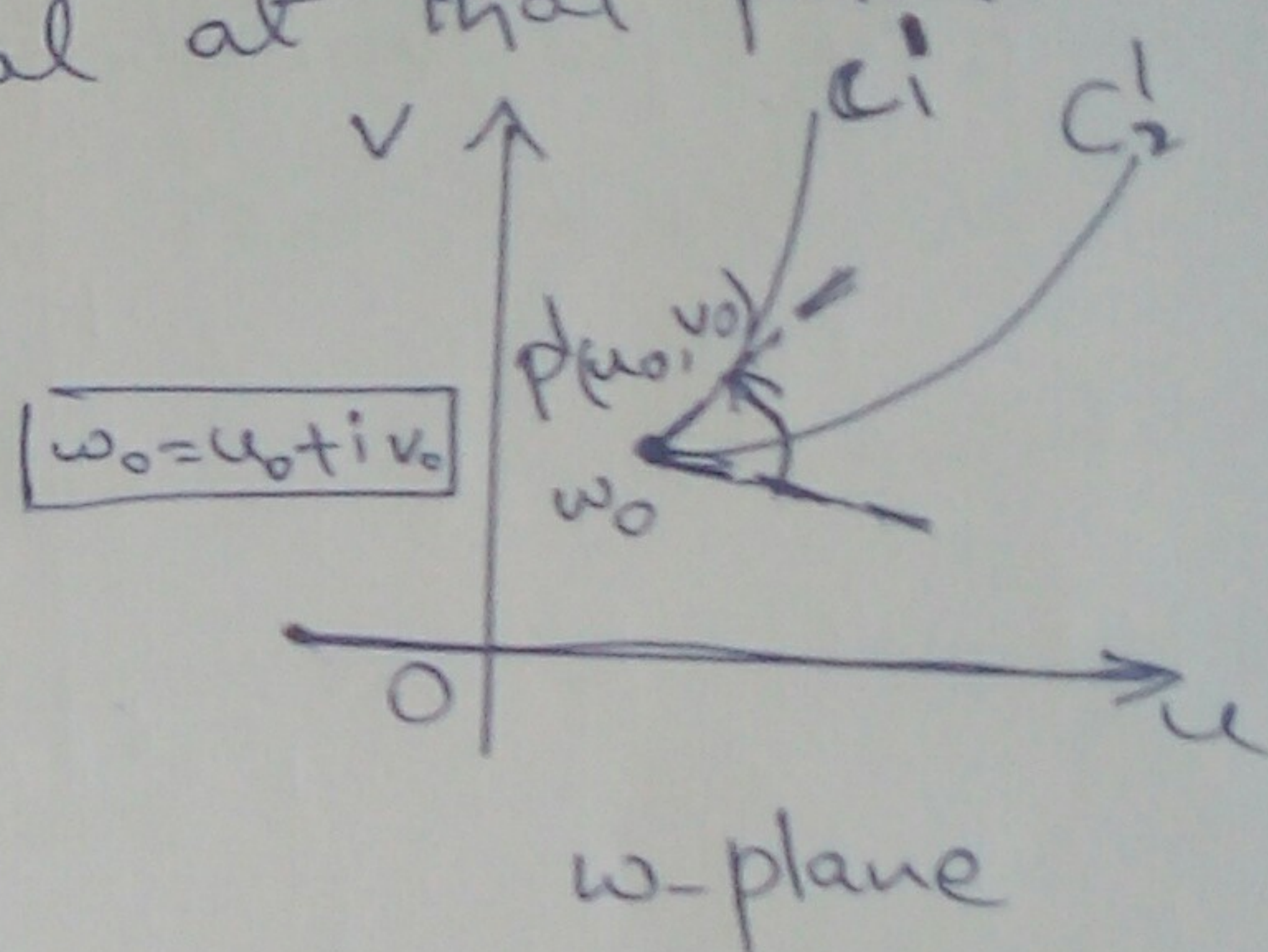
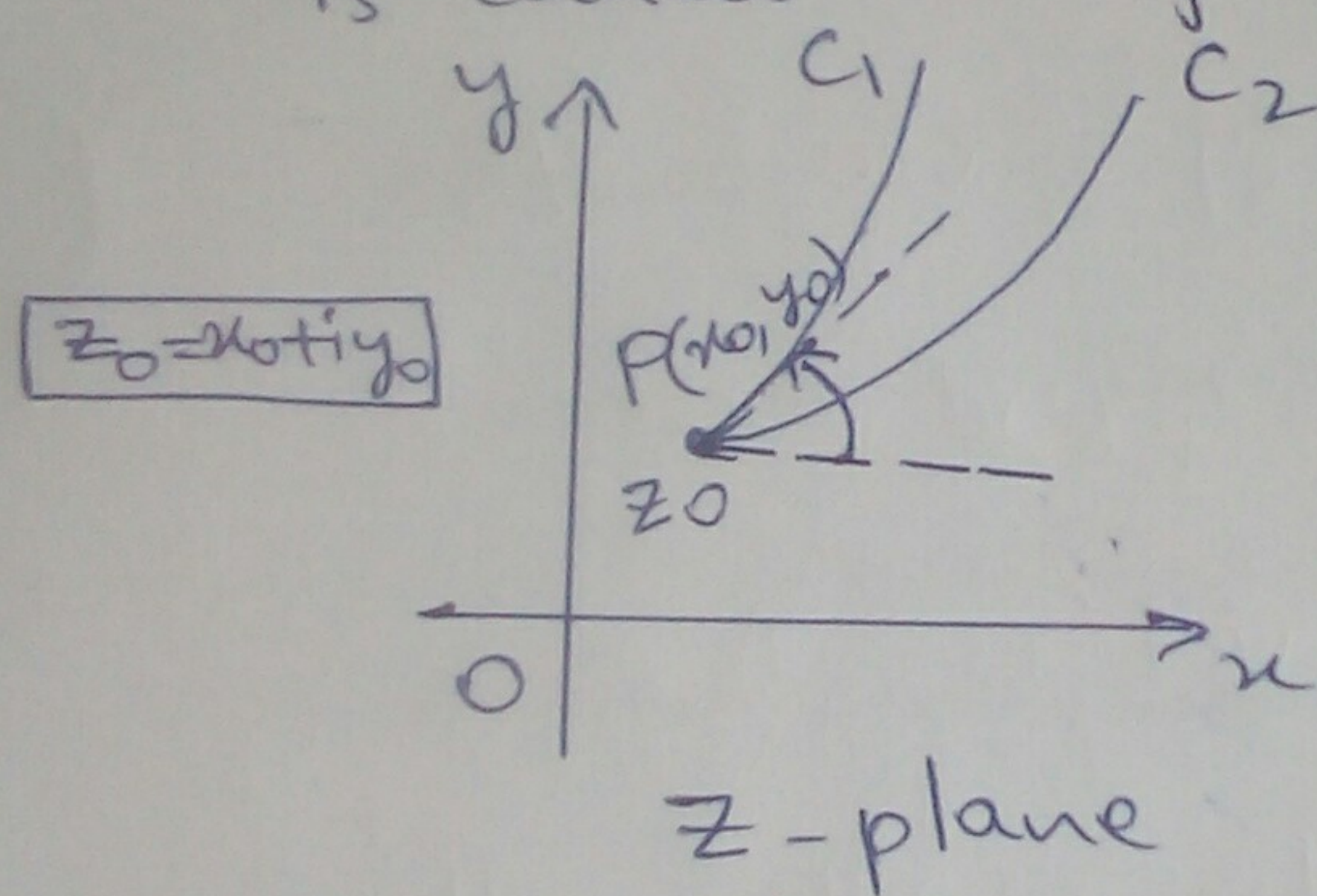


Module 2:

2.1 Conformal Mappings:

Def: A transformation that preserves angles between every pair of curves through a point, both in magnitude and sense (direction), is called conformal at that point.



Example:

$$\text{Let } w = z + 1 + i$$

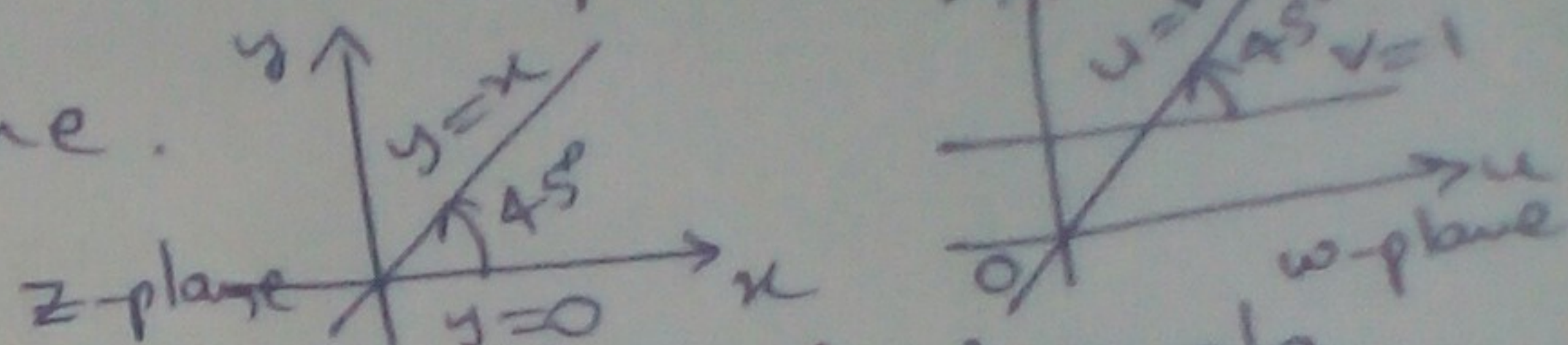
$$\text{i.e., } u + iv = (x + iy) + (1 + i)$$

$$\text{Then } u = x + 1 \text{ and } v = y + 1$$

Consider the straight line $y = x$.

$$\text{So, } u = x + 1, v = x + 1 \text{ and } u = v. \quad \text{hence}$$

Therefore, the line $y=x$ in z -plane is mapped onto $u=v$ on w -plane.



Taking $y=0$ and $y=x$. Then the angle between the lines $y=0$ and $y=x$ is 45° .

And the images are $v=1$ and $v=u$. clearly the angle between $v=1$ and $v=u$ is also 45° .

Therefore, the transformation $w = z + 1 + i$ preserves the angle and the direction (sense).

Hence this transformation is conformal.

Note: sufficient conditions for a transformation $w=f(z)$ to be conformal are

- (i) $f(z)$ is analytic
- and (ii) $f'(z) \neq 0$.

Some Standard transformations

Translation: The transformation $w = z + C$, where C is a complex constant, represents a translation.

Let $z = x + iy$, $w = u + iv$ and $C = a + ib$

given $w = z + C$. i.e., $u + iv = (x + iy) + (a + ib)$

$\Rightarrow u = x + a$ and $v = y + b$.

Hence the image of any point $P(x, y)$ in the z -plane is mapped ~~on to~~ onto the point $P(x+a, y+b)$ in the w -plane. Similarly every point in the z -plane is mapped onto the w -plane.

If we assume that the w -plane is superimposed on the z -plane, we observe that the point (x, y) and hence figure is shifted by distance $|c| = \sqrt{a^2 + b^2}$ in the direction of c (i.e., translated by the vector representing c).

Hence this transformation transforms a circle into an equal circle. Also the corresponding regions in the z -plane and w -plane will have the same size, shape and orientation.

Examples:

1. What is the region of the w -plane into which the rectangular region in the z -plane bounded by the lines $x=0, y=0,$

$x=1$ and $y=2$ is mapped under the transformation $w = z + (2-i)$

Sol: Given $w = z + (2-i)$

$$\text{i.e., } u+iv = (x+iy) + (2-i)$$

$$\Rightarrow u = x+2 \text{ and } v = y-1$$

given boundary lines are

$$x=0$$

$$y=0$$

$$x=1$$

$$y=2$$

transformed boundary lines are

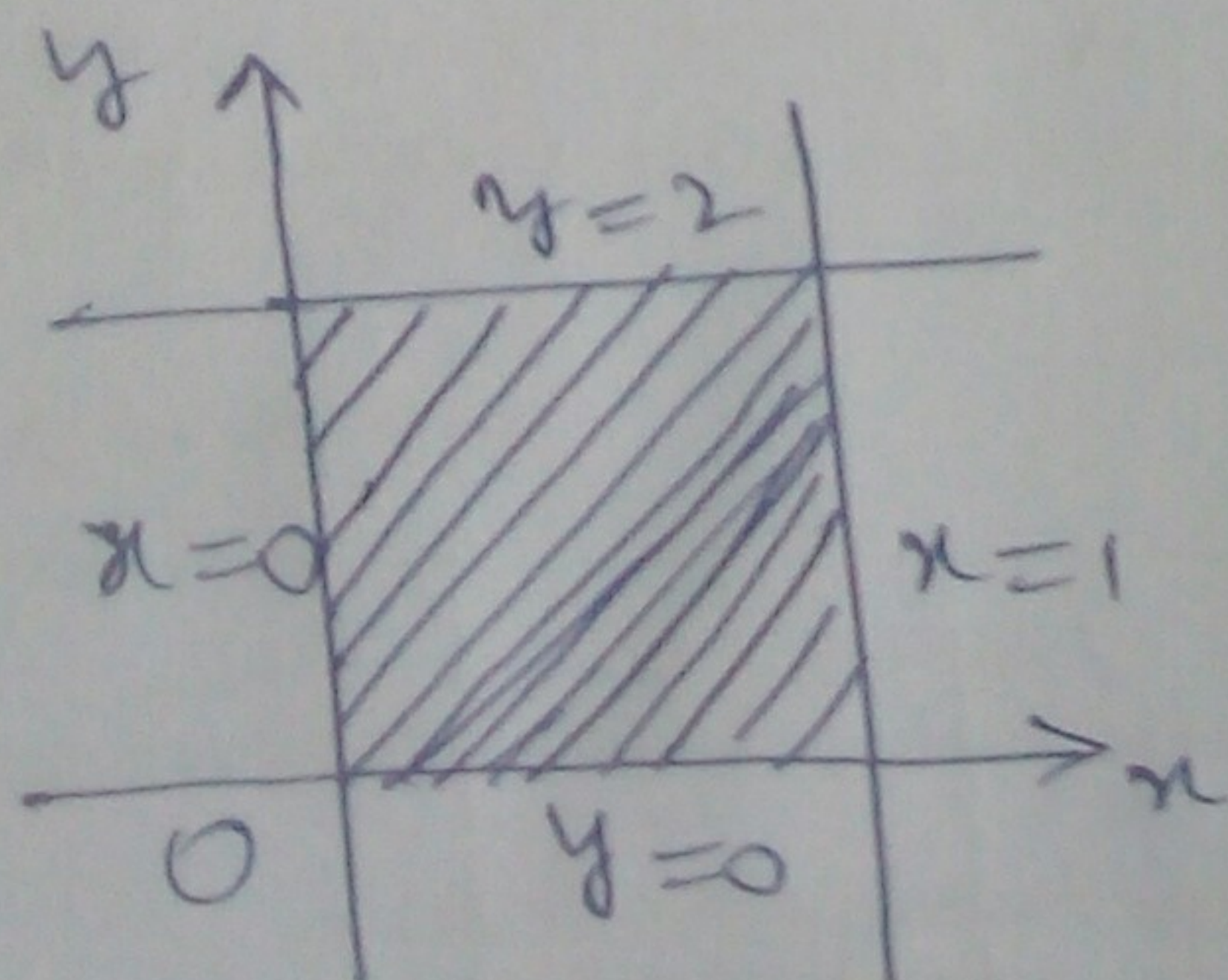
$$u = 0+2 = 2$$

$$v = y-1 = -1$$

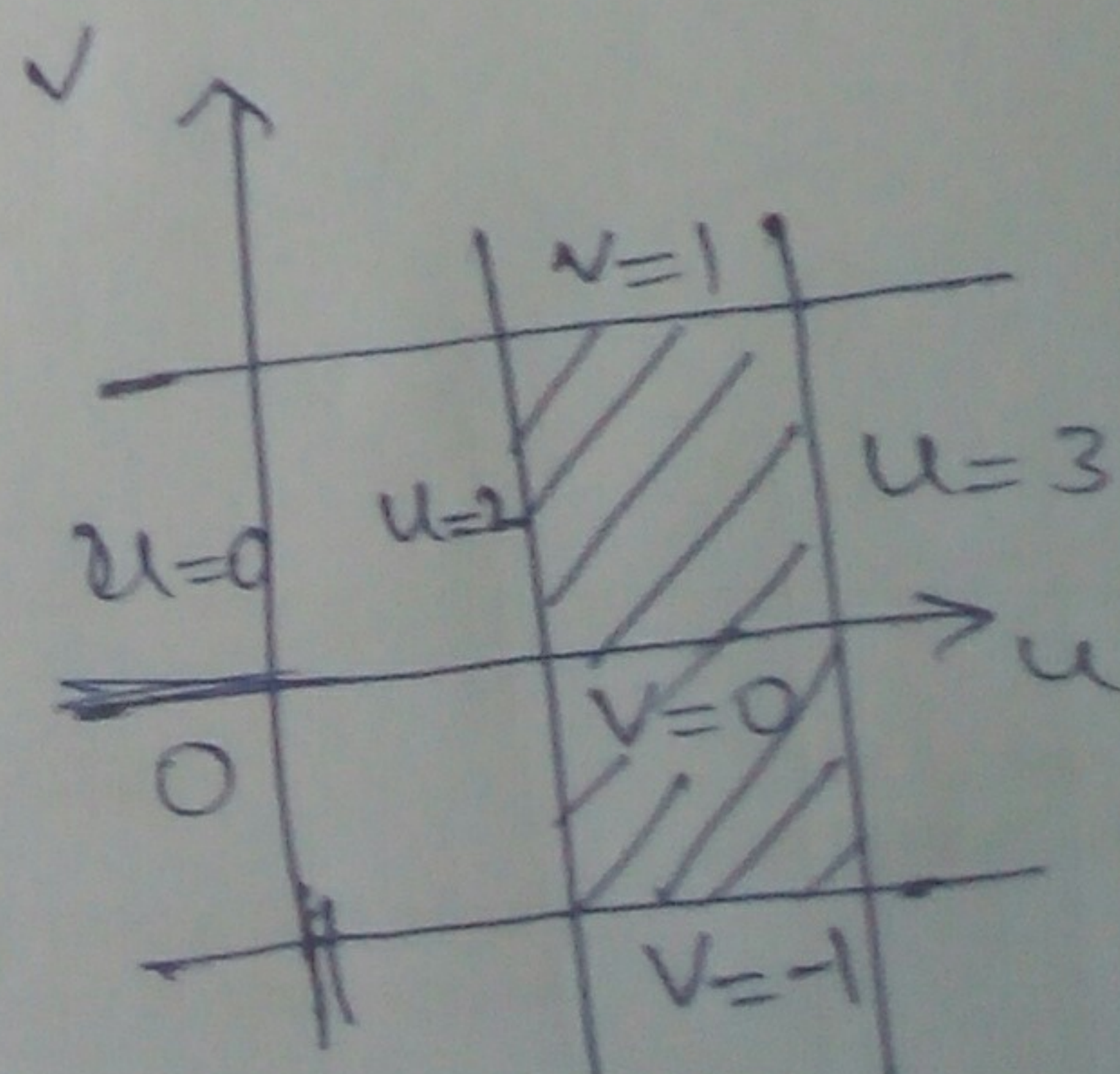
$$u = 1+2 = 3$$

$$v = 2-1 = 1$$

Hence the lines $x=0, y=0, x=1$ and $y=2$ are mapped into the lines $u=2, v=-1, u=3$ and $v=1$ respectively which form a rectangle in the w -plane.



z -plane



w -plane.

2. Find the image of the circle $|z|=1$ by the transformation $w = z + 2 + 4i$.

Sol: Given $w = z + (2 + 4i)$

i.e., $u + iv = (x + iy) + (2 + 4i)$

$$\Rightarrow u = x + 2 \text{ and } v = y + 4$$

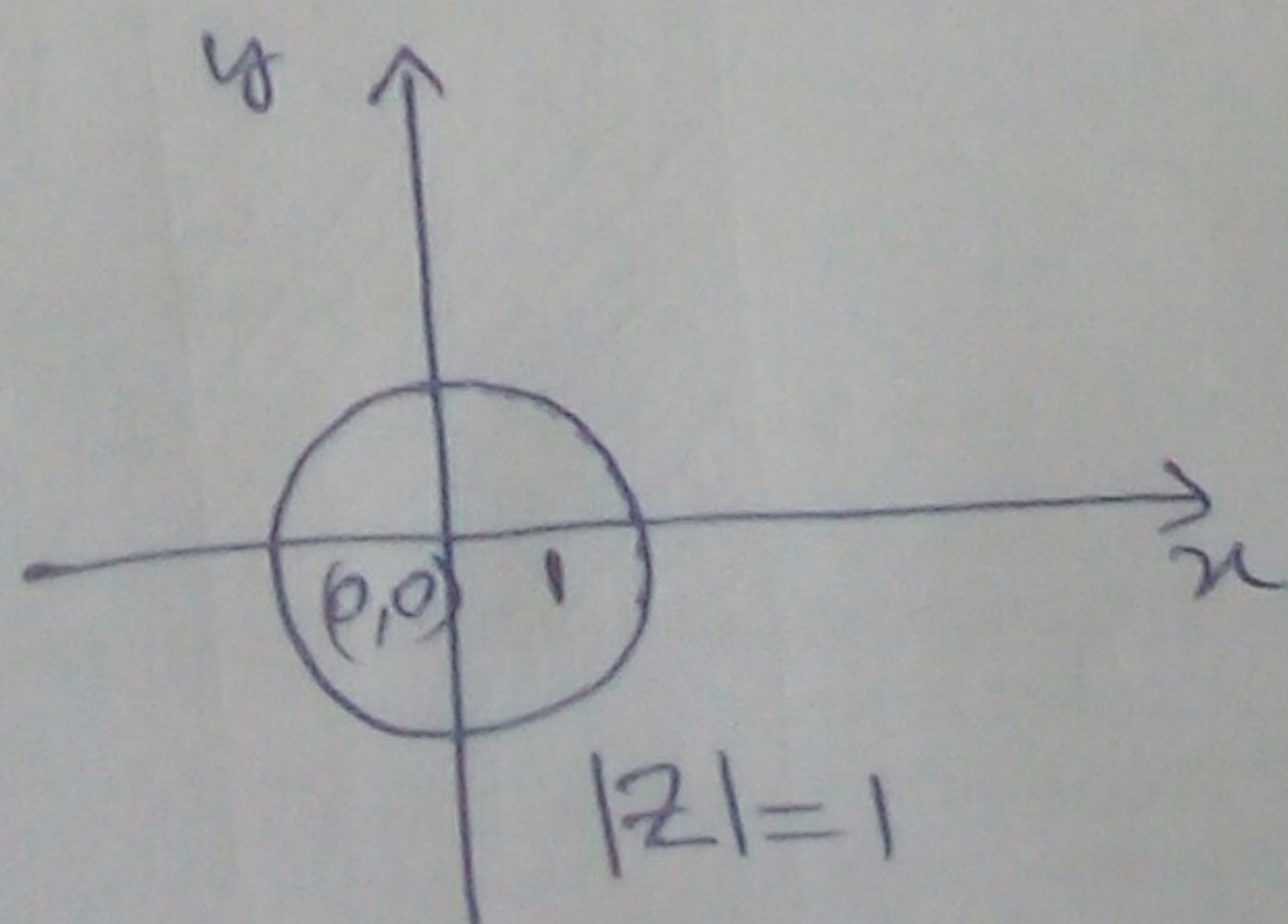
$$\text{or, } x = u - 2 \text{ and } y = v - 4.$$

Also, given $|z|=1$

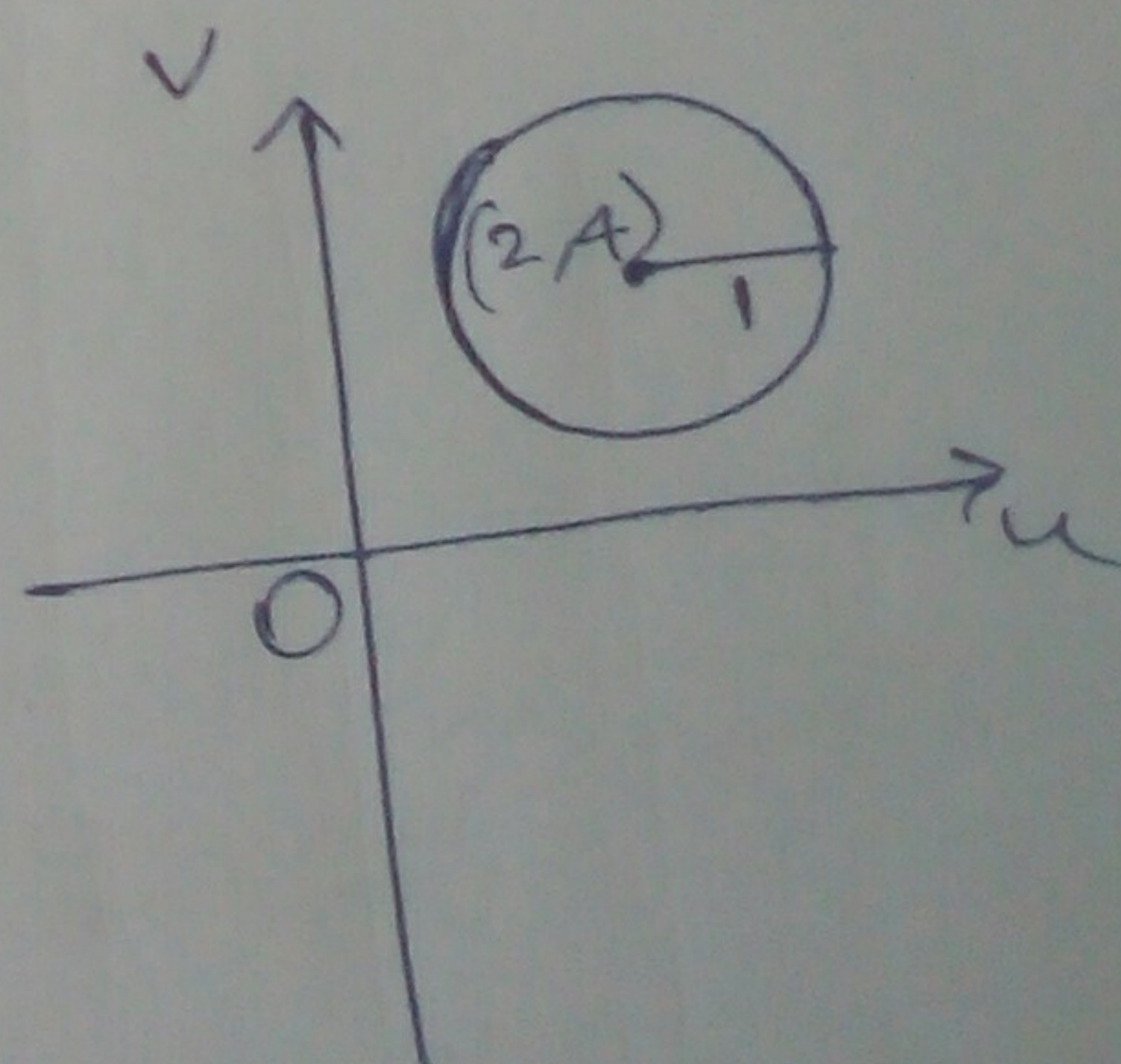
i.e., $x^2 + y^2 = 1$

$$\Rightarrow (u-2)^2 + (v-4)^2 = 1$$

Hence the circle $x^2 + y^2 = 1$ in z -plane is mapped into $(u-2)^2 + (v-4)^2 = 1$ in w -plane which is also a circle with centre $(2, 4)$ and radius 1.



z -plane



w -plane.